

1. PROJECT OBJECTIVE

This project involved building two interactive Power BI dashboards to analyze credit card customer behavior and transaction performance. The goal was to translate raw financial data into actionable business intelligence covering customer demographics, spending patterns, revenue trends, and profitability.

The dashboards were designed to be fully interactive, allowing stakeholders to filter and explore data dynamically across the following dimensions:

- Gender
- Age Group
- Income Group
- Week Number
- Expense Category

2. DATASETS USED

Two datasets were used in this project, joined through a common key to enable cross-dataset filtering.

Dataset 1 — Customer Data

Contains customer-level demographic and profile information:

- Client Number
- Gender
- Customer Age
- Annual Income
- Card Category
- Education Level

Dataset 2 — Credit Card Transaction Data

Contains weekly transaction activity and financial metrics:

- Weekly transaction records
- Total transaction amount and volume
- Interest earned
- Expense type
- Card usage details

3. DATA MODELING

The two datasets were connected using Client_Num as the primary key.

Relationship Type: One-to-Many

Direction: Customer → Transaction

This setup allowed customer attributes such as gender, age group, and income level to dynamically filter all transaction-related visuals in real time.

4. DAX CALCULATIONS

Calculated Columns

The following columns were derived to support segmentation and filtering:

- Age Group — Categorizes customers into defined age bands
- Income Group — Segments customers by income tier (Low / Medium / High)
- Week Number — Extracted from date for week-level trend analysis

Measures

The following measures were created to drive KPI cards and visual calculations:

- Total Revenue
- Total Transactions
- Total Interest Earned
- Current Week Revenue
- Previous Week Revenue
- Week-over-Week (WoW) Growth %
- Average Transaction Value
- Revenue Contribution %

Together, these measures enabled weekly performance monitoring, profitability tracking, and customer behavior analysis.

5. DASHBOARD 1 — CUSTOMER REPORT

Objective

This report focuses on customer demographics, segmentation, and revenue contribution by customer profile.

KPI Cards

- Total Customers
- Average Income
- Total Revenue
- Total Transactions

Visuals Included

- Revenue Trend by Gender
- Customers by Age Group
- Customers by Income Group
- Gender Distribution
- Customers by Marital Status
- Customer Segment Matrix (Income Group × Age × Card Category)

Key Insights

Weekly Revenue Trends

Weekly revenue showed fluctuations across the reporting period, with notable spikes during high-activity weeks. Male customers consistently generated a higher share of revenue compared to female customers. Week-over-Week Growth % analysis helped pinpoint periods of both positive momentum and contraction.

Customer Demographics

The 26–35 age group represented the largest share of the customer base, followed by the 36–45 group — together forming the dominant customer segment. Male customers accounted for slightly over 50% of the total customer count. Medium and high-income customers were the most active in terms of both engagement and revenue contribution.

Revenue Contribution

High-income customers generated significantly higher average revenue per customer compared to lower-income segments. Premium card holders, despite being a smaller portion of the base, contributed a disproportionately large share of total revenue and showed higher average transaction values and transaction frequency.

Customer Behavior

Middle-aged customer segments drove the majority of transaction activity. Certain clusters within the customer base contributed revenue well above average, indicating the presence of high-value customer segments. Revenue concentration varied noticeably across card categories and income groups.

6. DASHBOARD 2 — TRANSACTION REPORT

Objective

This report focuses on transaction-level behavior, spending category analysis, and weekly revenue performance.

KPI Cards

- Total Revenue
- Total Transactions
- Total Interest Earned
- Week-over-Week Growth %
- Current Week Revenue
- Previous Week Revenue

Visuals Included

- Revenue by Payment Method (Chip / Swipe / Online)
- Revenue by Expense Type
- Revenue and Interest Earned by Gender
- Revenue by Card Category
- Transaction Volume Trend
- Transaction Performance Matrix

Key Insights

Expense Category Analysis

Bills, Entertainment, Fuel, and Grocery emerged as the top revenue-generating expense categories. Notably, high transaction volume did not always correlate with high revenue — some categories showed elevated transaction counts but lower per-transaction value, while others generated strong revenue from fewer, higher-value transactions.

Transaction Volume Trends

Transaction volume generally tracked in line with revenue trends during peak weeks. Certain weeks saw an increase in transaction count paired with a drop in average transaction value, pointing to a shift toward smaller, more frequent purchases. Revenue spikes were more closely linked to higher-value transactions than to increased transaction frequency alone.

Card Category Performance

Premium card holders outperformed standard card holders on revenue contribution and average transaction value. Standard card categories drove higher raw transaction counts but contributed lower revenue per customer. This contrast highlights the outsized profitability impact of the premium card segment.

Profitability

Interest earned scaled proportionally with revenue across major customer segments. High-value customers contributed significantly more to both transaction revenue and interest income. Customers with higher transaction frequency also tended to show stronger overall profitability contribution.

7. INTERACTIVE FEATURES

Both dashboards were built with a strong emphasis on interactivity to support self-serve analysis:

- Slicers for Gender, Age Group, Income Group, Week Number, and Expense Category
- Dynamic cross-filtering between all visuals on each report page
- Drill-down hierarchy within matrix visuals for granular exploration
- Secondary axis on trend charts for comparing revenue and volume simultaneously
- Tooltips for additional detail on hover

8. CHALLENGES & RESOLUTIONS

Challenge 1 — Data Relationship Mapping

Establishing a clean, accurate relationship between the customer and transaction datasets was critical. Ensuring the join via Client_Num produced correct filtering behavior across both report pages required careful validation of the model structure and relationship direction.

Challenge 2 — Power BI Resource Limitation Errors

Using high-cardinality fields in certain visuals triggered resource limitation warnings from Power BI. This was resolved by:

- Replacing raw field references with optimized DAX measures
- Removing unnecessary columns from the visual query
- Redesigning affected visuals using summarized aggregations instead of row-level detail